

→ IDE (Integrated Development Environment)
vs code / Cursor / Tare / Antigravity

→ For running JS, we need Node.js

Process of Pulling code from Github

```
git config --global user.name "Rajat"
```

```
git config --global user.email "rajatkumarshri04@gmail.com"
```

```
git cd tab
```

```
git clone <repolink https> [Cloning repo from github]
```

```
git status (checking status)
```

```
git add. (tracking changes and update it) (green all the files after tracking)
```

```
git commit -m "first commit" (giving name to the version)
```

```
git status (now saying to push)
```

```
git push
```

For running locally

cd tab

npm run dev
(node package manager)

For Deployment

Vercel

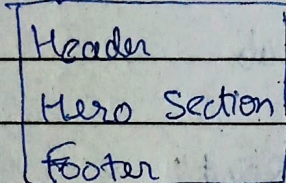
For Brand Selection

Name
Nameflix website
(domain availability
also be there)

* As Node Modules are heavy, so, we not push it to github (.gitignore file is there which ignore the modules)

* In public folder, there is images

In website



Ctr + F

↓
Finding any text
in code

In deployment -

After make changes in the local code, and push changes to github with push, Automating changes updated to Vercel

GitHub (All github project known as Repo (Repository))

- Pushed code to the cloud • Code is stored at remote level
- We can restore it any times if we lost the code (Backup)
- Version control & Switch between them
- Multiple persons can work at same time

Next JS → SEO friendly

Brandmark → website for logos / brands

Dribbble → for themes

Lovable → is an online ai tool that build websites without any dependencies (with no storage occupied)

- but had limited control • we can't scale our business
- Security threat

* The online editors (IDE) does not give direct option to download the code, to get the code in our

Online system, we can connect github, then clone the code from Github.

- Run the command `npm i` (it install node modules, without it cannot run the website) dependencies

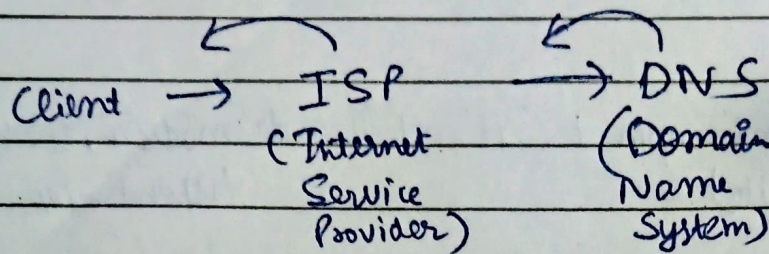
- `npm run dev` is used for running the code.

With n8n

→ we can make workflows (in sequence), it automate tasks

[with

Q How internet works?



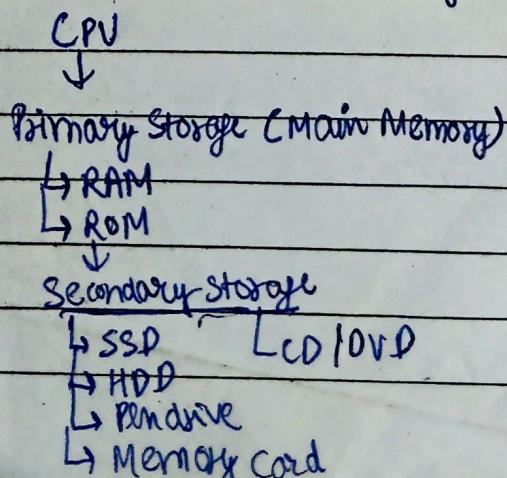
* Client type a url on Browser but browser doesn't know about where the URL is located, the request will go to Internet Service Provider then ISP send request to DNS then DNS find that the particular domain is registered or not then the particular IP address of the server is sent back to ISP then it will get back to the client

Rem
[Whether you search IP address or the URL basically the same thing]

Memory Hierarchy

↓
Arrangement of different type of memory in computer based on speed, cost and storage capacity.

- The closer the memory is to the CPU, the faster it is but it usually stores less data. The farther away, the slower it is, but it can store huge amount of data.



1) Primary Storage

↓
is the memory that the CPU can access directly

① RAM (Random Access Memory) → Can be read/write the data

- temporary (volatile) memory used to store programs and data that are currently running
- Expensive
- When you turn off the computer, everything in RAM is deleted
- Ex: When you open the Chrome, Chrome is loaded in RAM → Shutting down → Chrome is removed from RAM

② ROM (Read-Only Memory) → data is read-only

- Permanent Memory (firmware/BIOS)
- It stores the instruction to start the computer (Store Startup instruction)
- Data in ROM is not deleted when the power is off (Non-Volatile)

ROM: Ex: When you press power button, computer first read instructions from ROM to begin the boot process

RAM: Ex: GTA V is stored on SSD

- You click GTA V icon
- Computer copies GTA V from SSD to RAM with loader
- CPU reads it from RAM and runs the game

CPU is very fast but an SSD or HDD is much slower

② Secondary Storage

→ used to save data permanently

→ It keeps your files even when computer is turned off

→ CPU can't execute programs directly from it. Programs must be copied to RAM

① SSD (Solid State Drive)

- Permanent Storage • Very fast

② HDD (Hard Disk Drive)

- Permanent Storage • Slower than HDD

③ Pen drive → Portable USB storage device used to transfer files b/w computers

Firmware

is a small permanent software stored in ROM or flash memory. It controls the basic function of hardware device.

Ex BIOS Basic Input/output system

It is a type of ~~firmware~~ firmware stored on the computer motherboard. Used to start the computer. It is responsible for ~~BIOS~~ Booting the computer.

- 1) When the computer turns on, CPU first looks for the BIOS
- 2) BIOS ~~start~~ wakes up and begins checking the computer
- 3) It check → is the RAM working, keyboard connected, is the SSD/HDD detected, CPU working. This checking process is POST (Power on Self Test)
- 4) After checking hardware, BIOS first looks for the OS on the SSD/HDD
- 5) BIOS loads the Windows boot program. Windows starts loading into RAM. Now Windows takes control of computer
- 5) After Windows finishes loading, you see the desktop & open apps like Chrome or Word